

Seat No:

SET	S
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AUTONOMOUS STATE EXAMINATION BOARD**Btech Computer Scie Engg DBMS****Subject: Data Structure · Pattern: CBCS Pattern**

Day & Date: Thursday, September 17, 2026 **Duration:** 180 Minutes **Max. Marks:** 70 Marks

Instructions for Candidates:

1. Q.1 is compulsory. Mention the question paper set (**S**) clearly on top of the answer booklet.
2. Figures to the right indicate full marks assigned to each question.
3. Assume suitable data wherever necessary and state your assumptions explicitly.
4. Use of programmable calculators or unauthorized electronic communication devices is strictly prohibited.
5. Standard University Marking Scheme

Q.1 Choose the correct alternatives for the following questions. [14 Marks]

1. Which algorithm is utilized by operating systems to safely avoid deadlocks when processes declare their maximum resource claims in advance?
 - a) Banker's Algorithm
 - b) Peterson's Algorithm
 - c) Lamport's Bakery Algorithm
 - d) Dekker's Algorithm
2. What is the minimum number of scalar multiplications required to multiply a chain of matrices with dimensions: A1 (10x30), A2 (30x5), A3 (5x60)?
 - a) 4,500
 - b) 2,700
 - c) 18,000
 - d) 3,000
3. In a B+ Tree of order p, where are the actual data record pointers (or records) stored?
 - a) Only in the internal nodes
 - b) Only in the leaf nodes
 - c) Uniformly in both leaf and internal nodes
 - d) At the root node only
4. In open addressing with linear probing, what phenomenon occurs when filled hash slots form continuous clusters, degrading search performance?
 - a) Secondary Clustering
 - b) Primary Clustering
 - c) Hash Overflow

- d) Perfect Hashing
5. Which normal form is based on the concept of full functional dependency and eliminates partial dependency on candidate keys?
- a) First Normal Form (1NF)
 - b) Second Normal Form (2NF)
 - c) Third Normal Form (3NF)
 - d) Boyce-Codd Normal Form (BCNF)
6. Belady's Anomaly where allocating more page frames results in more page faults can occur in which page replacement algorithm?
- a) Optimal Page Replacement (OPT)
 - b) Least Recently Used (LRU)
 - c) First-In First-Out (FIFO)
 - d) Least Frequently Used (LFU)
7. Which of the following sorting algorithms is inherently stable and operates with $O(n \log n)$ worst-case time complexity?
- a) Quick Sort
 - b) Heap Sort
 - c) Merge Sort
 - d) Selection Sort
8. Which algorithm finds the single-source shortest paths in a directed graph containing edges with negative weights (assuming no negative weight cycles)?
- a) Dijkstra's Algorithm
 - b) Bellman-Ford Algorithm
 - c) Prim's Algorithm
 - d) Kruskal's Algorithm
9. In database transaction management, which ACID property guarantees that concurrent execution of transactions results in a state equivalent to serial execution?
- a) Atomicity
 - b) Consistency
 - c) Isolation
 - d) Durability
10. Which relational algebra operation selects rows from a relation that satisfy a specified predicate or condition?
- a) Projection (π)
 - b) Selection (σ)
 - c) Cartesian Product ($\times$)
 - d) Natural Join ($\bowtie$)

11. What is the worst-case time complexity of searching an element in an AVL Tree with n nodes?
- a) $O(1)$
 - b) $O(\log n)$
 - c) $O(n)$
 - d) $O(n \log n)$
12. Which CPU scheduling algorithm gives the minimum average waiting time for a given set of processes?
- a) First-Come First-Served (FCFS)
 - b) Shortest Job First (SJF / Preemptive SRTF)
 - c) Round Robin (RR)
 - d) Priority Scheduling
13. Which locking protocol prevents cascading aborts in concurrent transaction execution?
- a) Basic Two-Phase Locking (2PL)
 - b) Strict Two-Phase Locking (Strict 2PL)
 - c) Conservative Two-Phase Locking
 - d) Graph-based Tree Locking Protocol
14. Which normal form is based on the concept of full functional dependency and eliminates partial dependency on candidate keys?
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 - d) Boyce-Codd Normal Form (BCNF)
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